Practice Question Bank: Numerical Problems in Speech Technology

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Section 1: Sampling and Frame Theory

1. **Question:** A speech signal is sampled at 32 kHz. Calculate the exact number of samples contained in a 25 ms analysis window.
   Solution:
   $$\text{Samples} = \text{Sampling Rate (Hz)} \times \text{Time (s)}$$
   $$\text{Samples} = 32,000 \times 0.025 = 800 \text{ samples}$$

2. **Question:** If a frame contains 512 samples and the sampling rate is 16,000 Hz, what is the duration of the frame in milliseconds?
   Solution:
   $$\text{Duration} = \frac{\text{Samples}}{\text{Sampling Rate}} = \frac{512}{16,000} = 0.032 \text{ seconds}$$
   $$0.032 \times 1000 = 32 \text{ ms}$$

3. **Question:** An audio file consists of 480,000 samples. How many 20 ms non-overlapping frames can be extracted if the sampling rate is 48 kHz?
   Solution:
   $$\text{Total duration} = \frac{480,000}{48,000} = 10 \text{ seconds}$$
   $$\text{Number of frames} = \frac{10\text{s}}{0.020\text{s}} = 500 \text{ frames}$$

Section 2: Neural Codecs and Residual Vector Quantization (RVQ)

4. **Question:** A neural codec uses 10 RVQ layers. If the codebook size at each layer is 1024, how many total bits are required to represent a single audio frame?
   Solution:
   $$\text{Bits per layer} = \log_2(1024) = 10 \text{ bits}$$
   $$\text{Total bits} = 10 \text{ layers} \times 10 \text{ bits/layer} = 100 \text{ bits}$$

5. **Question:** A Speech LLM processes audio at 50 frames per second (Hz). If it uses 8 RVQ codebooks and a flattened autoregressive strategy, what is the total token sequence length for a 10-second audio clip?
   Solution:
   $$\text{Sequence Length} = \text{Time} \times \text{Frame Rate} \times \text{Number of Codebooks}$$
   $$10 \times 50 \times 8 = 4,000 \text{ tokens}$$

6. **Question:** To achieve a specific bit-depth, Standard VQ requires a codebook of size 2^{24}. If we switch to RVQ with 3 layers to achieve the same bit-depth, what is the size of the codebook at each individual layer?
Solution:
$$\text{Total bits} = 24. \text{ For 3 layers, bits per layer} = \frac{24}{3} = 8 \text{ bits}$$
$$\text{Codebook size per layer} = 2^8 = 256 \text{ entries}$$

Section 3: CNN Dimensionality and Sequence Reduction

7. **Question:** A raw audio input has $16,000$ samples. It passes through a 1D-CNN feature extractor with Kernel $K = 100$, Stride $S = 50$, and Padding $P = 0$. Calculate the output temporal dimension O.
Solution:
$$O = \lfloor \frac{I - K + 2P}{S} \rfloor + 1$$
$$O = \lfloor \frac{16,000 - 100 + 0}{50} \rfloor + 1 = \lfloor \frac{15,900}{50} \rfloor + 1 = 318 + 1 = 319$$

8. **Question:** You wish to reduce a 16 kHz signal to a 50 Hz latent representation. If you use a single 1D-CNN layer, what should the Stride (S) be set to?
Solution:
$$S = \frac{\text{Input Rate}}{\text{Output Rate}} = \frac{16,000}{50} = 320$$

Section 4: Vector Quantization Math

9. **Question:** Given a latent vector $z = [2, 3]$ and a codebook $C = \{c_1 = [1, 1], c_2 = [3, 4]\}$. Which centroid will z be quantized to based on Euclidean distance?
Solution:
$$d(z, c_1)^2 = (2 - 1)^2 + (3 - 1)^2 = 1 + 4 = 5$$
$$d(z, c_2)^2 = (2 - 3)^2 + (3 - 4)^2 = 1 + 1 = 2$$
Since $2 < 5$, the winning centroid is c_2.

10. **Question:** In an RVQ system, the first layer quantizes $z = [10, 10]$ to $c_1 = [8, 8]$. What is the residual vector r that is passed to the second quantization layer?
Solution:
$$r = z - c_1 = [10 - 8, 10 - 8] = [2, 2]$$

Section 5: Evaluation Metrics

11. **Question:** An ASR system transcribes a 10-word reference sentence. The output has 1 substitution, 1 deletion, and 1 insertion. Calculate the Word Error Rate (WER).
Solution:
$$\text{WER} = \frac{S + D + I}{N} = \frac{1 + 1 + 1}{10} = \frac{3}{10} = 30\%$$

12. **Question:** If a speech tokenizer uses a vocabulary of 2048 discrete tokens, how many bits of information does each token represent?
Solution:
$$\text{Bits} = \log_2(2048) = 11 \text{ bits}$$